Polynomial Kernel

a b

Remember : $(a \cdot b) = (a_1, a_2)(b_1, b_2) = (a_1 b_1 + a_2 b_2) = a_1 b_1 + a_2 b_2 + 2a_1 b_1 a_2 b_2$

$(a \cdot b + 1) = (a \cdot b) + 2a \cdot b + 1 = (a_1 b_1 + a_2 b_2) = a_1 b_1 + a_2 b_2 + 2a_1 b_1 a_2 b_2 + 2(a_1 b_1 + a_2 b_2) + 1$

$= a_1 b_1 + a_2 b_2 + 2a_1 b_1 a_2 b_2 + 2a_1 b_1 + 2a_2 b_2 + 1$

Where : $\varphi(x) = (x_1, \ldots$